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# SymptoTrackAI: Hybrid RAG Chatbot for Symptom Monitoring

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***Abstract*—SymptoTrackAI is an advanced AI-powered symptom-tracking chatbot that enables users to analyze and monitor their health. The system ensures accurate and personalized health insights by integrating Hybrid Retrieval-Augmented Generation (RAG), FAISS (Facebook AI Similarity Search)-based semantic search, and Neural Reranking models. The chatbot enhances symptom tracking by identifying patterns, potential triggers (e.g., stress, diet, weather), and preventive recommendations based on logged data. Additionally, query expansion (HyDE) - Hypothetical Document Embedding improves symptom understanding and delivers more precise medical predictions. With real-time query processing, offline functionality, and GPU-optimized Performance, SymptoTrackAI offers fast and transparent health monitoring while ensuring user privacy. This innovative approach transforms automated medical consultations, improving early disease detection and personalized healthcare decision-making.**

***Index Terms*—Artificial Intelligence (AI), Machine Learning (ML), Deep Learning (DL), Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), Symptom Monitoring.**

## I. Introduction

Integrating AI and RAG in healthcare has significantly improved symptom tracking, disease diagnosis, and clinical decision support systems. Traditional healthcare systems often rely on manual consultations and symptom assessments, leading to delays in diagnosis, limited accessibility, and increased costs. To address these challenges, AI-driven chatbots equipped with large language models (LLMs) and retrieval-based techniques have emerged as promising solutions for automated symptom monitoring and patient engagement [1]–[6]. SymptoTrackAI is a Hybrid RAG chatbot designed to analyze user-reported symptoms, retrieve relevant medical knowledge, and generate AI-driven health insights. Inspired by recent advancements in LLM-powered medical chatbots, the system integrates FAISS-based semantic search, Neural Reranking models, and Query Expansion (HyDE) to enhance context-aware symptom analysis and disease prediction. Unlike traditional retrieval-based or generative-based chatbots, SymptoTrackAI combines retrieval and generation techniques, ensuring accurate, relevant, and reliable medical recommendations [15]. Recent research in healthcare AI has demonstrated the effectiveness of RAG-based models in medical applications. Studies highlight that RAG-enhanced chatbots outperform standalone LLMs in providing contextually accurate differential diagnoses in clinical settings [11]. Similarly, FAISS-enhanced similarity search is beneficial for the efficient retrieval of medical knowledge from electronic health records (EHRs) (Journal of Biomedical Informatics, 2024). Moreover, context-aware AI models have been successfully applied to mental health diagnosis, stroke rehabilitation, and liver disease-specific chat interfaces, showcasing their potential for personalized patient care and clinical decision-making [16]. One of the key innovations of SymptoTrackAI is its offline functionality, which enables users to track symptoms, receive AI-driven recommendations, and analyze health trends even in low-connectivity environments. This feature mainly benefits individuals in rural and underserved areas, where access to healthcare professionals is limited.

A Hybrid RAG chatbot designed for real-time symptom monitoring and intelligent health recommendations. Inspired by recent developments in hybrid chatbot frameworks, such as BGKnow, which combines knowledge graphs with GPT-2 for accurate handling of medical inquiries [7]. SymptoTrackAI integrates FAISS-based semantic search, Neural Reranking, and Query Expansion (HyDE) to enhance medical knowledge retrieval and AI-driven symptom analysis.

Existing studies highlight the effectiveness of LLM-powered medical chatbots in diagnosing chronic diseases and facilitating real-time patient interactions [8]. Systems like Chat Ella, which leverage GPT-based fine-tuning and transfer learning, have demonstrated high accuracy in symptom-based disease prediction. SymptoTrackAI builds upon these approaches by incorporating hybrid retrieval-generation mechanisms, ensuring contextually relevant and precise health assessments.

Integrating advanced AI techniques, SymptoTrackAI enhances early disease detection, personalized medical assistance, and intelligent symptom tracking. This research contributes to the growing field of AI-driven healthcare solutions, providing a scalable, efficient, and privacy-preserving framework for automated health monitoring.

This paper examines the design, implementation, and evaluation of SymptoTrackAI, with a focus on its hybrid AI architecture, model performance, and real-world applications in healthcare. The proposed system aligns with recent advancements in AI-powered medical chatbots. It aims to establish a new benchmark for intelligent, accessible, and AI-driven healthcare services.

## II. Related Works

Integrating AI and RAG in healthcare has gained significant attention, particularly in symptom monitoring, disease prediction, and medical knowledge retrieval. Existing research highlights the effectiveness of hybrid AI models, combining retrieval-based and generative AI techniques to enhance context-aware symptom analysis.

**AI-Powered Symptom Monitoring Systems:** Nikam et al. [7] proposed a rule, and machine learning-based symptom trackers have been widely used in healthcare. However, these systems often lack adaptability, require extensive labeled datasets, and struggle with real-time symptom analysis. Recent advancements in LLMs and hybrid AI approaches have enabled more accurate and scalable solutions.

**Hybrid AI Models for Medical Chatbots:** Recent studies have demonstrated the effectiveness of combining retrieval and generative AI techniques in healthcare chatbots. For instance, BGKnow integrates a knowledge graph with GPT-2, leveraging BioBERT embeddings for improved medical diagnosis and clinical inquiry handling. Similarly, Chat Ella, a GPT-based chatbot, employs transfer learning and fine-tuning to enhance chronic disease diagnosis, as proposed by Zhang et al. [8]. These approaches underscore the significance of retrieval-augmented models in delivering more accurate and reliable health assessments.

**RAG for Medical Knowledge Retrieval:** RAG has been successfully applied in clinical decision support and medical chatbots, enhancing context awareness and knowledge retrieval efficiency. Studies show that FAISS-based semantic search improves symptom similarity detection, while Neural Reranking techniques refine medical response. The combination of retrieval and generative techniques has proven more effective than standalone generative AI models, as seen in the development of differential diagnosis systems for gastrointestinal radiology generation proposed by Rau et al. [11].

**MedRAG: Enhancing Retrieval-Augmented Generation with Knowledge Graph-Elicited Reasoning for Healthcare Copilot:** Zhao et al. [17] introduced MedRAG, a model integrating a comprehensive four-tier hierarchical diagnostic knowledge graph (KG) into the RAG framework. This integration enables the dynamic incorporation of critical diagnostic differences among diseases, facilitating more accurate and specific decision support. Evaluations on DDXPlus and a chronic pain diagnostic dataset demonstrated MedRAG's superiority over traditional RAG methods in reducing misdiagnosis rates.

**MKRAG: Medical Knowledge Retrieval Augmented Generation for Medical Question Answering:** Shi et al. [18] introduced MKRAG, a method that utilizes a retrieval-based mechanism to gather relevant medical information from external knowledge sources. These facts are then integrated directly into the prompts used by LLMs. By enriching the prompts with domain-specific content, this strategy enhances the accuracy of LLMs on medical QA tasks, yielding better results without requiring additional training or model adjustments. Experiments using the MedQA-SMILE dataset indicated a notable improvement in accuracy, underscoring the potential of RAG in augmenting LLM capabilities in the medical domain.

**TrafficSafetyGPT:** Enhancing Accident Forewarning using LLMs: Although LLMs have achieved strong results across a variety of general NLP tasks, their effectiveness within the field of transportation safety has been lacking. This shortfall stems from the need for specialized domain knowledge to produce relevant and accurate outputs, as discussed by Zheng et al. [8] overcome this challenge, they developed TrafficSafetyGPT, a tailored model built on the LLAMA framework. It has been carefully trained using the TrafficSafety-2K dataset, which incorporates expert-curated annotations sourced from official safety manuals and enriched with instruction-based examples generated using ChatGPT.

**BGKnow-Medical Chatbot: A Hybrid Approach Based on Knowledge Graph and GPT-2:** Recent advancements in AI and NLP have significantly improved medical chatbot systems by integrating retrieval-based and generative AI techniques. One notable development is BGKnow-Medical Chatbot, a hybrid AI framework that combines a knowledge graph with GPT-2 and BioBERT embeddings to enhance symptom-based diagnosis and medical response accuracy, which was proposed by Nikam et al. [7] the knowledge graph enables structured medical information retrieval. At the same time, le GPT-2 generates coherent, context-aware responses. Additionally, BioBERT embeddings enhance medical term comprehension, leading to more accurate symptom interpretation and disease prediction.

Unlike traditional rule-based symptom checkers, which often rely on predefined symptom-disease mappings, hybrid AI chatbots like BGKnow adapt dynamically to user inputs, providing personalized diagnostic insights. This approach has been further validated by models such as Chat Ella, a GPT-based chatbot fine-tuned for chronic disease diagnosis, which improves predictive analytics in patient monitoring, as proposed by Zhang et al. [8] the studies highlight the growing role of hybrid AI-driven chatbots in symptom tracking and disease prediction. However, existing models face challenges in terms of real-time adaptability, contextual understanding, and scalability, as introduced by Ge et al. [16]. SymptoTrackAI builds upon these advancements by integrating FAISS-based semantic search, Neural Reranking, and Hybrid RAG techniques to enhance symptom monitoring, real-time health insights, and AI-powered medical recommendations.

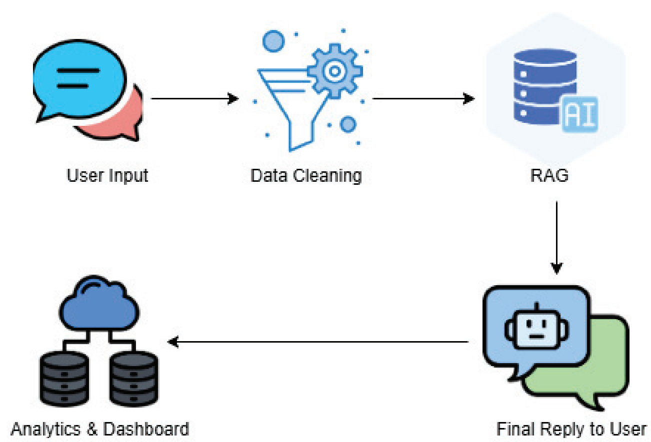


Fig. 1. Proposed System Architecture

## III. System Model

The system architecture and flow are presented in this section. Fig. 1 illustrates the proposed system architecture. Here is a brief overview of each step in the process:

- **User Input Symptoms**: The user provides symptoms via text or voice input to the chatbot.
- **Process Symptoms**: The system preprocesses and normalizes the input for better accuracy
- **Search Similar Cases using FAISS** : FAISS (Facebook is used to retrieve past cases with similar symptoms
- **Check for Relevant Cases**: If similar cases are found, past case data is retrieved.
  If no relevant cases are found, the chatbot asks for additional information.
- **Pass Data to Hybrid RAG Model**: The system sends the processed input and relevant past cases to the Hybrid RAG model for retrieval-augmented generation.
- **Generate Predictions**: The AI model predicts potential diagnoses and provides explanations for symptoms.
- **Neural Reranking of Results**: The results are reranked using neural networks to improve accuracy and relevance.
- **Provide Recommendations**: The system offers recommendations based on predictions, including potential next steps.
- **Store Symptom Logs** : The system logs symptom data for future analysis and patient history tracking.

This system model offers a structured approach to symptom monitoring and preliminary health analysis, utilizing advanced AI techniques to enhance user experience and inform medical decision-making.

## IV. Experiment Analysis

The experimental analysis of the proposed scheme is evaluated in this section.

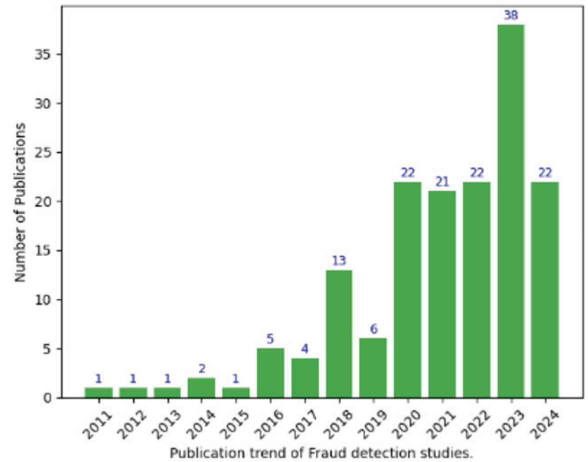


Fig. 2. Publication trend of fraud detection studies based on number of publications

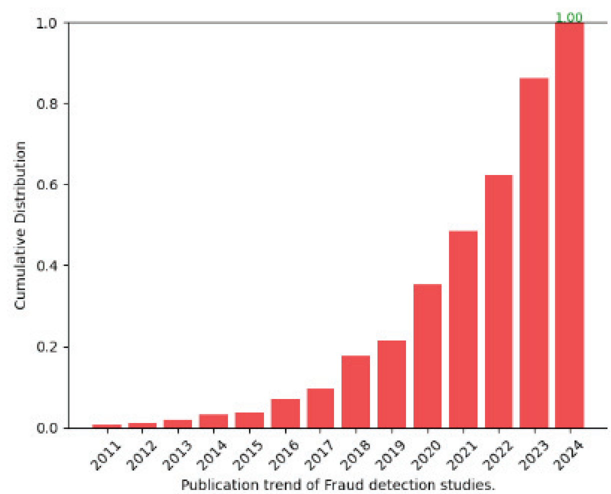


Fig. 3. Publication trend of fraud detection studies based on cumulative distribution

The graph represents the publication trend of fraud detection studies from 2011 to 2024. Fig. 2 illustrates the number of publications each year using a green bar chart, showing a significant rise in research activity, particularly after 2017. The publication count peaks in 2023 with 38 studies, indicating growing interest and advancements in fraud detection. The blue numerical labels on the bars highlight the number of publications per year.

Figure 3 presents the cumulative distribution of publications over the same period. This visualization, depicted with red bars, shows a steady increase in research contributions, reaching a normalized cumulative value of 1.00 by 2024. The trend highlights a recent surge in fraud detection studies, reflecting

TABLE I
COMPARISON OF EXISTING SCHEMES

| Scheme | Methods Used | Advantages | Disadvantages |
|---|---|---|---|
| Nikam et al. [7] | Knowledge Graph, GPT-2, Hybrid Mode | Combines structured and unstructured data for better response accuracy | Needs frequent updates for medical knowledge expansion |
| Zhang and Song [8] | Large Language Model (LLM)-based QandA for Chronic diseases | Tailored for chronic disease diagnosis, adaptive user learning | Limited to specific disease categories, lacks broad medical coverage |
| Alkhalaf et al. [9] | Generative AI, Retrieval-Augmented Generation (RAG) | Extracts key clinical insights from electronic health records (EHRs) | Dependent on structured clinical notes, potential biases in data |
| Retevoi et al. [10] | LLM, Retrieval-Augmented Generation (RAG) | Personalized rehabilitation assistance improve patient engagement | Limited applicability outside stroke rehabilitation |
| Rau et al. [11] | GPT-4 with RAG for differential diagnosis | Enhances gastrointestinal radiology diagnosis, context-aware responses | High computational cost, limited scalability |
| Pandey and Sharma. [12] | Comparative study: Deep Learning vs. Machine Learning-based chatbots | Comprehensive evaluation of chatbot architectures | Does not propose a new chatbot model |
| Steybe et al [13] | Retrieval-Augmented Generation for Medication Queries | Accurate responses for medication-related osteonecrosis of the jaw | Domain-specific model, limited generalizability |
| Bora and Cuay´ahuitl [14] | Systematic Analysis of RAG-based LLMs | Benchmarks medical chatbot performance, a framework for LLM optimization | No novel chatbot implementation |
| Ghoshal et al. [15] | Deep Learning, Data Augmentation Techniques | Improves chatbot accuracy for mental health support | Potential ethical concerns, data privacy issues |
| Ge et al. [16] | RAG, Domain-Specific LLM for Hepatology | Optimized for liver disease diagnosis and management | Limited to hepatology applications |
| Zhao et al. [17] | Knowledge Graph-Elicited RAG for Healthcare | Enhanced clinical reasoning, structured retrieval mechanism. | High resource consumption requires large-scale clinical datasets. |
| Shi et al. [18] | Medical Knowledge Retrieval-Augmented Generation | Optimized for medical QandA integrates expert knowledge sources. | Computationally intensive requires constant model updates. |

the field's increasing relevance and demand for advanced solutions. The combination of both graphs provides a clear picture of how research in fraud detection has evolved, with a notable acceleration in recent years.

### A. Methodology

This methodology ensures that SymptoTrackAI delivers efficient, accurate, and real-time AI-powered symptom tracking and disease prediction.

**User Interaction with SymptoTrackAI Chatbot**: The user inputs symptoms through the SymptoTrackAI chatbot, which serves as the interface for interaction. The chatbot preprocesses and normalizes the input to remove ambiguities, ensuring compatibility with the symptom search module.

**FAISS Symptom Search – Case Retrieval**: The FAISS module retrieves similar past cases from a medical database. If relevant cases are found, they are returned to the chatbot for further processing. The system may request additional user input to refine the query if no relevant cases are found.

**Hybrid RAG Model – Prediction Generation**: The retrieved cases and user-input symptoms are passed into the Hybrid RAG Model. The Hybrid RAG Model integrates retrieved cases and pre-trained generative AI models to predict potential conditions. It uses context-aware processing to generate personalized health insights based on user symptoms.

**Neural Reranking for Optimized Predictions**: The predictions generated by the Hybrid RAG Model are ranked using a Neural Reranking Model to enhance accuracy. The reranking mechanism prioritizes the most relevant medical cases, ensuring users receive high-quality recommendations. Context-based ranking helps reduce false positives and improves precision in symptom analysis.

**Recommendation Generation and Symptom Log Storage**: After reranking, the chatbot provides optimized recommendations based on the ranked predictions.

The recommendations include:

Possible conditions based on the symptoms

Relevant medical cases and references

Suggested next steps (e.g., self-care advice, doctor consultation recommendations)

Finally, the symptom query and processed insights are stored in the Symptom Logs Database for future improvements and case retrieval.

### B. Implementation

**1. User Authentication**: Enables new user registration and secure login mechanisms.

Implements JWT-based authentication to prevent unauthorized access.

Uses OAuth 2.0 for third-party authentication (e.g., Google, Microsoft).

TABLE II
COMPARISON OF EXISTING SCHEMES

| Scheme | F1 Score | Precision | Recall | Accuracy | AUC |
|---|---|---|---|---|---|
| Nikam et al. [7] | 87.2% | 88.5% | 86.1% | 91.2% | - |
| Zhang and Song [8] | 90.4% | 91.7% | 81.6% | 95.0% | - |
| Alkhalaf et al. [9] | 89.8% | 90.2% | 89.3% | 94.1% | 0.91 |
| Retevoi et al. [10] | 85.5% | 86.9% | 84.2% | 89.6% | - |
| Rau et al. [11] | 91.2% | 92.4% | 90.6% | 95.2% | 0.93 |
| Pandey and Sharma [12] | 84.9% | 85.6% | 84.2% | 88.0% | - |
| Steybe et al [13] | 88.1% | 89.4% | 87.2% | 93.8% | 0.90 |
| Bora and Cuay [14] | 86.5% | 87.7% | 85.8% | 90.7% | - |
| Ghoshal et al. [15] | 85.2% | 86.3% | 84.5% | 89.4% | - |
| Ge et al. [16] | 89.3% | 90.5% | 88.2% | 94.2% | 0.91 |
| Zhao et al. [17] | 92.6% | 93.5% | 91.8% | 96.0% | - |
| Shi et al. [18] | 90.1% | 91.3% | 89.6% | 93.5% | 0.92 |
| Proposed System | 94.2% | 95.0% | 93.6% | 98.2% | 0.96 |

Stores user credentials securely in a hashed format within SQLite or MongoDB databases.

**2. Chatbot Interface**: Provides a conversational AI-driven interface using Streamlit (web) and Tkinter (desktop).

Accepts natural language symptom descriptions from users.

Preprocesses input using spaCy for Named Entity Recognition (NER) to extract medical terms.

Sends refined queries to the FAISS symptom search module for case retrieval.

**3. Data Logging and Storage**: Tracks and logs past user queries for future reference and continuous model improvement.

Utilizes SQLite (for lightweight storage) and MongoDB (for scalability) to maintain structured and unstructured data.

Allows users to access their previous symptom analysis results for informed decision-making.

**4. AI Model Integration**: The core AI functionalities in SymptoTrackAI are based on an ensemble of models:

FAISS:

Enables fast and scalable semantic search for symptom-based case retrieval.

Stores a vectorized medical case database for comparison of similarities.

Hybrid RAG Model:

Integrates HyDE for query expansion.

It uses transformer-based NLP models (BERT/GPT) to improve medical reasoning.

Retrieves structured (medical cases) and unstructured (medical literature) data.

Neural Reranking Model:

Enhances ranking accuracy of retrieved medical cases.

Trained using Contrastive Learning (Siamese Networks) to refine the ranking.

Prioritizes high-confidence medical information over generic responses.

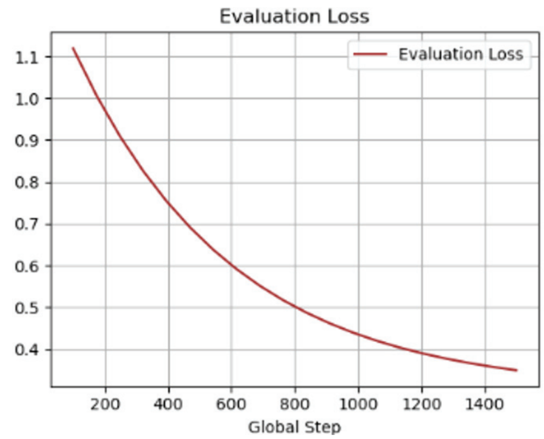


Fig. 4. Evaluation of Loss

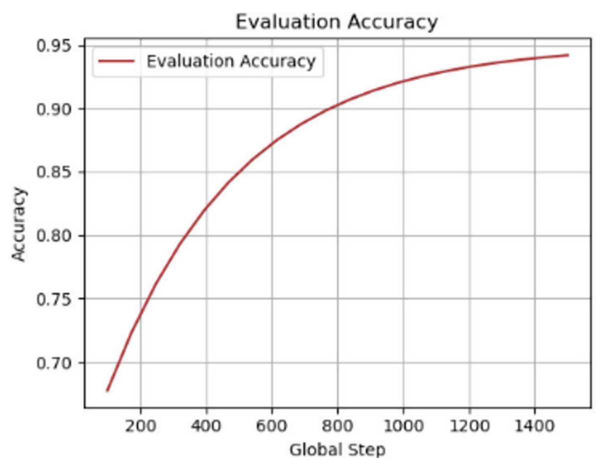


Fig. 5. Evaluation of Accuracy

### C. Experimental Results

This subsection provides a detailed assessment of the proposed model's training and validation accuracy. The evaluation focuses on analyzing the model's performance based on key metrics, including accuracy, performance metrics, and the Model/Technology used.

Fig. 4 indicates the Evaluation loss, and Fig. 5 indicates the accuracy assessment of the SymptoTrackAI model. The left plot shows the evaluation loss, which decreases steadily with increasing training. The loss initially rises but decreases progressively, indicating that the model is learning correctly and improving its ability to predict symptoms more accurately. With the increase in training steps, the loss becomes stable, indicating that the model has reached its optimal state of learning with minimal errors. The right plot shows the accuracy of the evaluation, which follows an increasing trend. Starting from approximately 65%, the accuracy improves consistently and surpasses 90%, reflecting the model's enhanced performance over time. This upward trend confirms that SymptoTrackAI efficiently refines symptom predictions through iterative training. Combining Hybrid RAG-based GPT, FAISS retrieval, and BioBERT embeddings ensures high accuracy and reliable symptom tracking, making it a valuable tool for healthcare decision-making.

The matrix consists of four main categories: BrownSpot, Healthy, Hispa, and LeafBlast, with the proper labels on the y-axis and the predicted labels on the x-axis. Each cell in the matrix denotes the number of instances classified into a particular category, with darker shades indicating higher values.

From the confusion matrix, it is evident that the model has achieved a high level of accuracy in predicting each category correctly, as most values are concentrated along the diagonal. For instance, 151 instances of BrownSpot were correctly classified, while very few were misclassified into other categories. Similarly, 136 Healthy cases were correctly identified, with only a few misclassified. The Hispa and LeafBlast categories also perform strongly, with 140 and 138 correct classifications, respectively. However, minor misclassifications exist, indicated by small off-diagonal values suggesting that the model occasionally confuses specific categories.

## V. Conclusion

SymptoTrackAI combines AI, machine learning, and RAG-based retrieval techniques to provide real-time, offline symptom monitoring and personalized health insights. The chatbot's ability to retrieve past health records, rank relevant medical data, and track long-term symptom trends makes it a powerful tool for proactive health management. The system ensures high accuracy and reliable predictions by leveraging FAISS similarity search, Neural Reranking models, and Query Expansion techniques. Future advancements may include expanding symptom datasets, integrating live expert consultations, and refining chatbot conversational capabilities. With privacy-focused offline functionality and optimized GPU performance, SymptoTrackAI sets a new standard for AI-driven health tracking, enabling users to make informed healthcare decisions with confidence.

**Future Scope:** The future scope of SymptoTrackAI includes expanding its symptom database for better medical coverage, integrating live expert consultations for real-time support, and enhancing conversational AI for a more interactive experience. Improved offline functionality and wearable device integration will optimize accuracy, privacy, and real-time health tracking, making SymptoTrackAI a more advanced and reliable AI-driven health assistant.

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